ELEG 5693 Wireless Communications
Course Syllabus

General Information: Instructor: Jingxian Wu | Office: Bell 3181
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Office Hour: Tu. Th. 11:00-12:00

Required Material:
- Textbook: "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005. ISBN: 978-0521837163".
- Software: Matlab with Signal Processing Toolbox

Reference:
- "Principles of Mobile Communication", G.L. Stuber, 2nd ed., Kluwer Academic, 2001.
- "Wireless Communications: principles and practice", T.S. Rappaport, 2nd ed., Prentice Hall, 2002.

Prerequisites: Signal and System, Probability and Random Process, Communication Theory
- Familiar with Matlab Programming
- Knowledge of linear time invariant system, Fourier series and transform, Laplace transform, time domain and frequency domain representation of signals, power spectrum density, and energy spectrum density of deterministic signals
- Knowledge of discrete and continuous random variable, probability mass function, probability density function, moments of random variables, moment generating function
- Knowledge of random process, wide sense stationary random process, auto-correlation function, power spectrum density of random signals
- Knowledge of digital modulation and demodulation
- Knowledge of optimum receiver design

Learning Objectives: To understand the theory of wireless propagation environment and wireless communication systems, to be able to design and analyze optimum receivers for various wireless communication in fading channel, to understand the theories and practice of diversity, multiple input multiple output (MIMO) system, space time coding, spread spectrum modulation, orthogonal frequency division multiple access, to be able to simulate and design wireless communication systems.

Grading:
- Test 1 (Chs. 0 – 3) 35%
- Test 2 (Chs. 4 – 9) 35%
- Homework 30%

- A: $90 \leq \text{grade} \leq 100$
- B: $80 \leq \text{grade} < 90$
- C: $70 \leq \text{grade} < 80$
- D: $60 \leq \text{grade} < 70$
- F: $0 \leq \text{grade} < 60$

- Due dates for homework and lab report will be strictly enforced. Late submission within one week after due date will receive a 20% grade deduction, and no credit if submitted after one week from the due date.
- If for some legitimate reason (sickness, death in the family, etc.), you cannot take an **exam** on the scheduled day, you must notify the instructor **prior** to the exam.

Online Resources:

- Course materials (Slides, Homework, Projects, References, etc) can be found at http://comp.uark.edu/~wuj/teaching/eleg5693/eleg5693.html
- Please check course website **at least once per week** for updates.

Academic Honesty:

Each University of Arkansas student is required to be familiar with and abide by the University's 'Academic Integrity Policy' which may be found at http://provost.uark.edu/
Students with questions about how these policies apply to a particular course or assignment should immediately contact their instructor.

Tentative Schedule:

- Week 1: Ch. 1 Overview of Wireless Communications (Lectures 1, 2)
- Week 1: Ch. 0 Math review (Lectures 3)
- Week 2: Ch. 0 Math review (Lecture 4, 5)
- Week 2: Ch. 2 Propagation and Noise (Lectures 6)
- Week 3: Ch. 2 Propagation and Noise (Lecture 7, 8, 9)
- Week 3: Ch. 3 Digital Modulation (Lecture 9)
- Week 4: Ch. 3 Digital Modulation (Lectures 10, 11, 12)
- Week 5: Ch. 4 Source and Channel Coding (Lectures 13, 14, 15)
- Week 6: Ch. 5 Equalization (Lecture 16, 17)
- Week 6: Ch. 6 Diversity and MIMO (Lecture 18)
- Week 7: Ch. 6 Diversity and MIMO (Lecture 19)
- Week 7: Ch. 7 Multicarrier Modulation (Lectures 19, 20, 21)
- Week 8: Ch. 7 Multicarrier Modulation (Lecture 22)
- Week 8: Ch. 8 CDMA (Lectures 22, 23)
- Week 8: Ch. 9 Ad hoc wireless network (Lecture 24)

Project Schedule:

(Optional for online students)

- Project 1: Simulator of Fading
- Project 2: BER of 8PSK in Fading
- Project 3: Convolutional Coding and Decoding
- Project 4: Space-Time Block Code
- Project 5: Orthogonal Frequency Division Multiplexing

The above schedule is subject to change without prior notice.